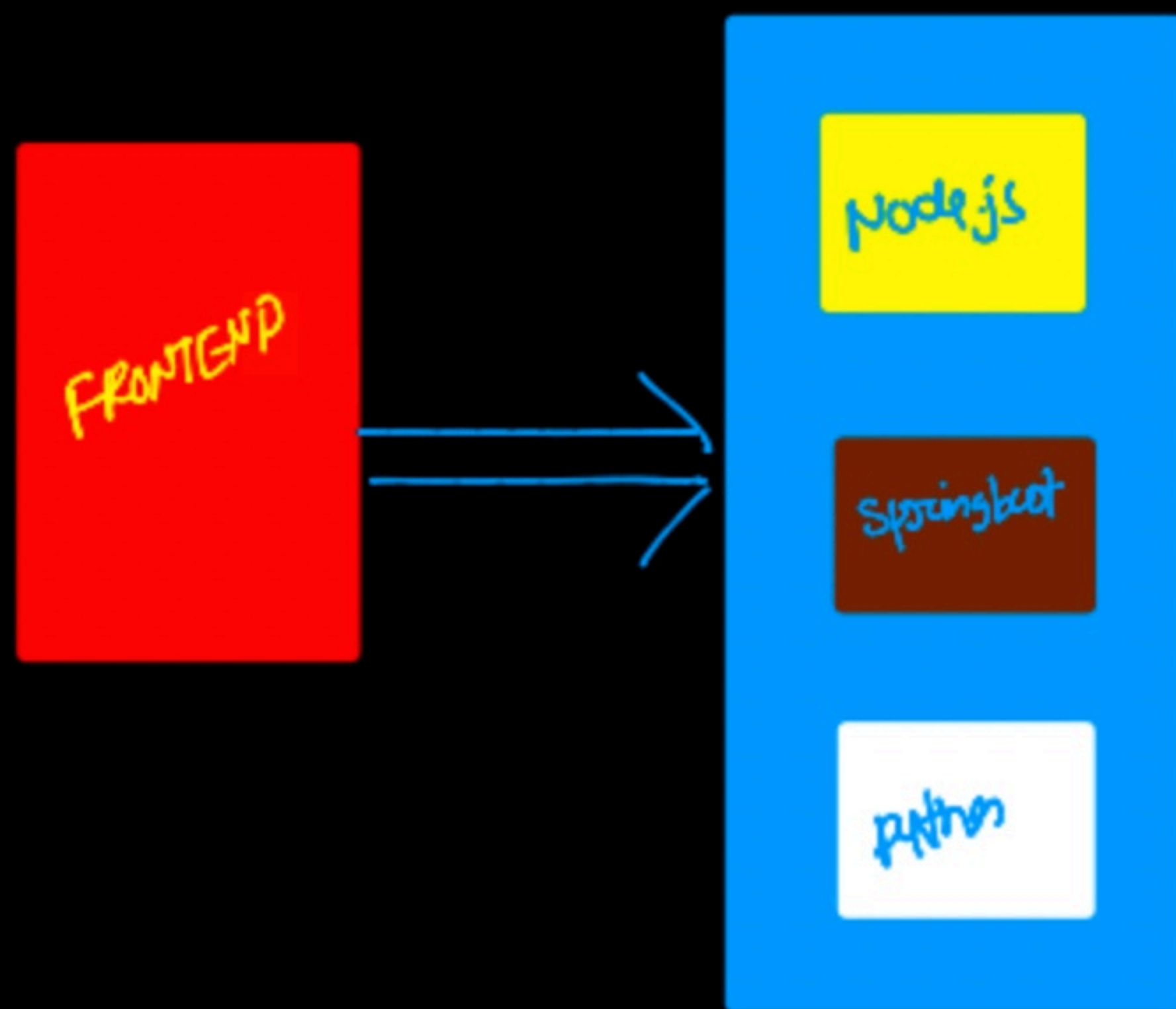


①

Docker solved "it works on my M/C" by packaging app and dependency into container



What if (you are just using Docker)

- A container crashes at 2am.
- Traffic suddenly jumps 10x?
- You need 5 instance of service
- One container run on another VM.
- You deploy a new version without downtime.

② # What is orchestration?

↳ Orchestration means automatically managing

- many containers
- Scaling → ?
- N/w → ?
- self healing

Docker
↓
Runs one
containers

Orchestration
↓
manages
many
containers

③ # The container orchestration problems

Problem	Example
① Container crashes	→ who restart it?
② Scaling	→ How to run 10 copies.
③ load balancing	→ Which container get traffic
④ Service discovery	→ How does service-A find service-B
⑤ Zero time deploy	→ How to update without outage
⑥ Multiple MIC setup	→ Docker work per MIC


KUBERNETES COMES AS HERO

66 Kubernetes is a container platform that manages containers for you

↳ automates → deployment → Scaling → healing → H/W and management of containerized applications

DOCKER			Kubernetes	
①	Run containers	→	manage containers	
②	Single host focus	→	multi node clusters	
③	Manual scaling	→	Auto scaling	
④	Manual re-start	→	self healing	
⑤	Manual H/W	→	Build-in service discovery	

⑤

Kubernetes architecture

↳ it works as a cluster model where cluster means

Control Plane + Worker Node

⇓
[Think like brain
which decides
what should
happen]

⇓
They will
do
the job

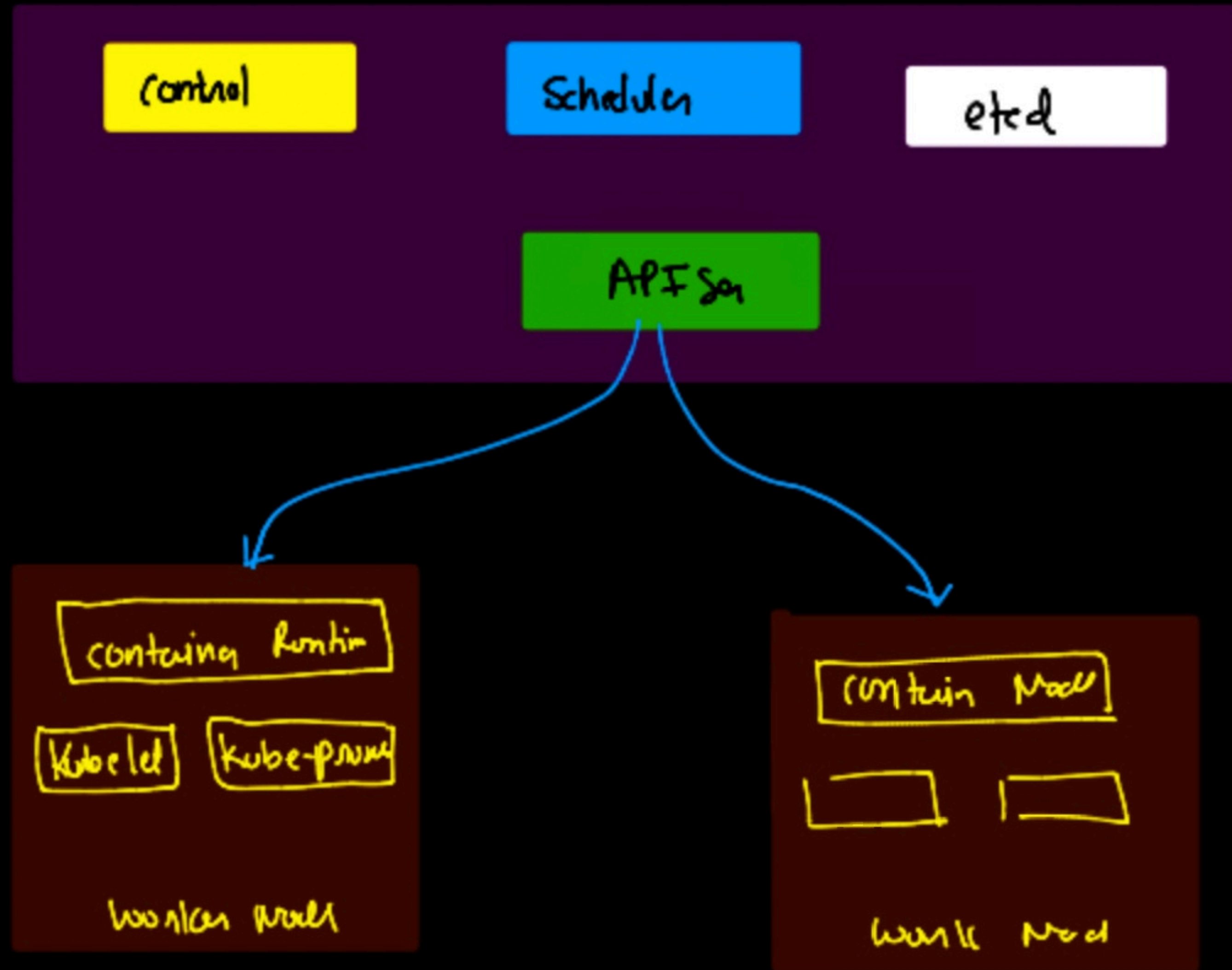
Control Pane

control
manager

Scheduler

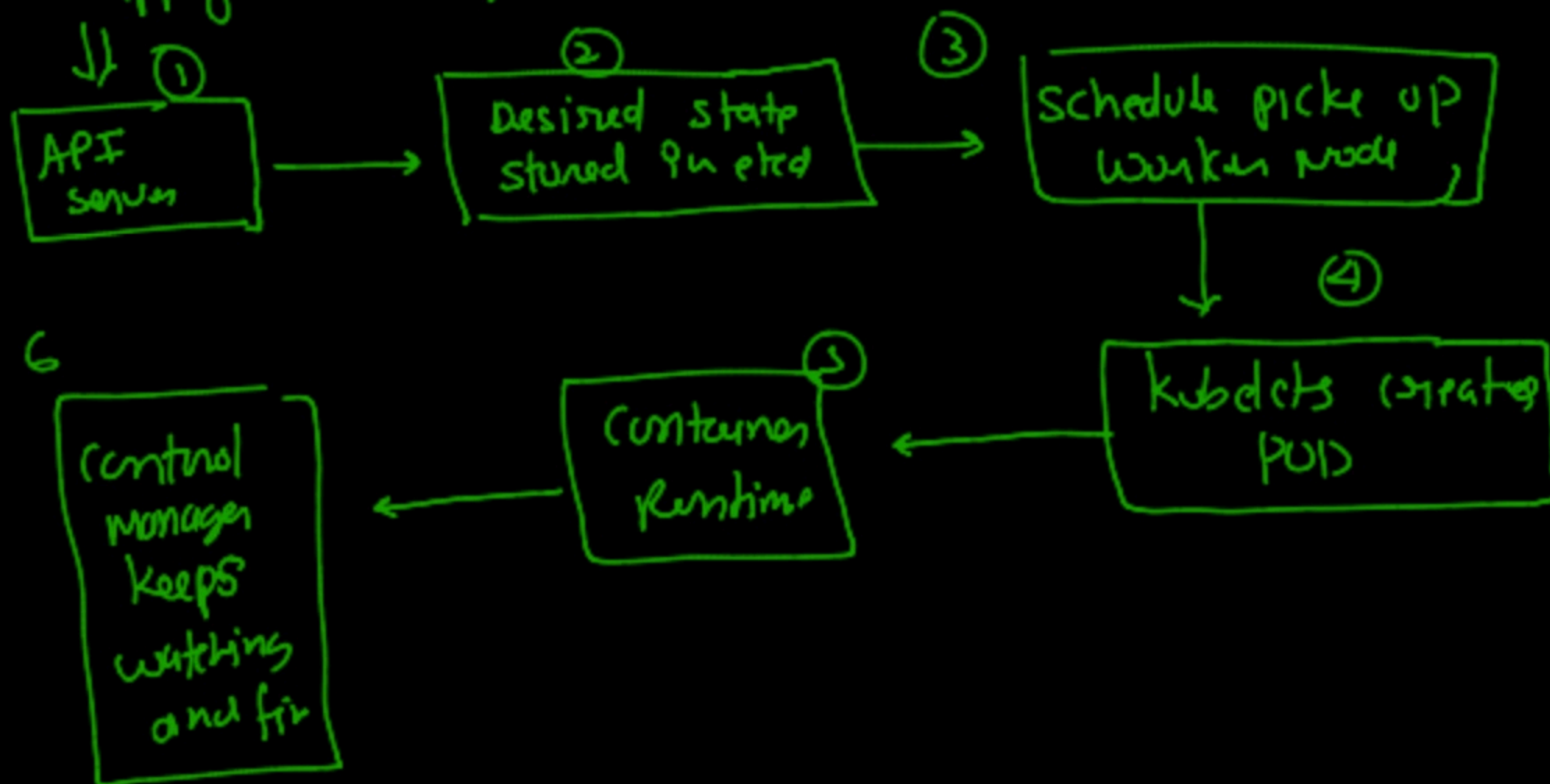
etc d

API Server



- ① Control Plane → decides and manages
- ② Worker Node → runs your applications

Simple eg → `kubectl apply -f app.yaml`



Setting up k8s locally

follow session

① PODS
↓

① the actual running app

② PODS where your application live and run → kind of HOME of your app

② Deployment

↳ its a component → which runs your app continously
↳ if you wants more copies, Deployment handle it.